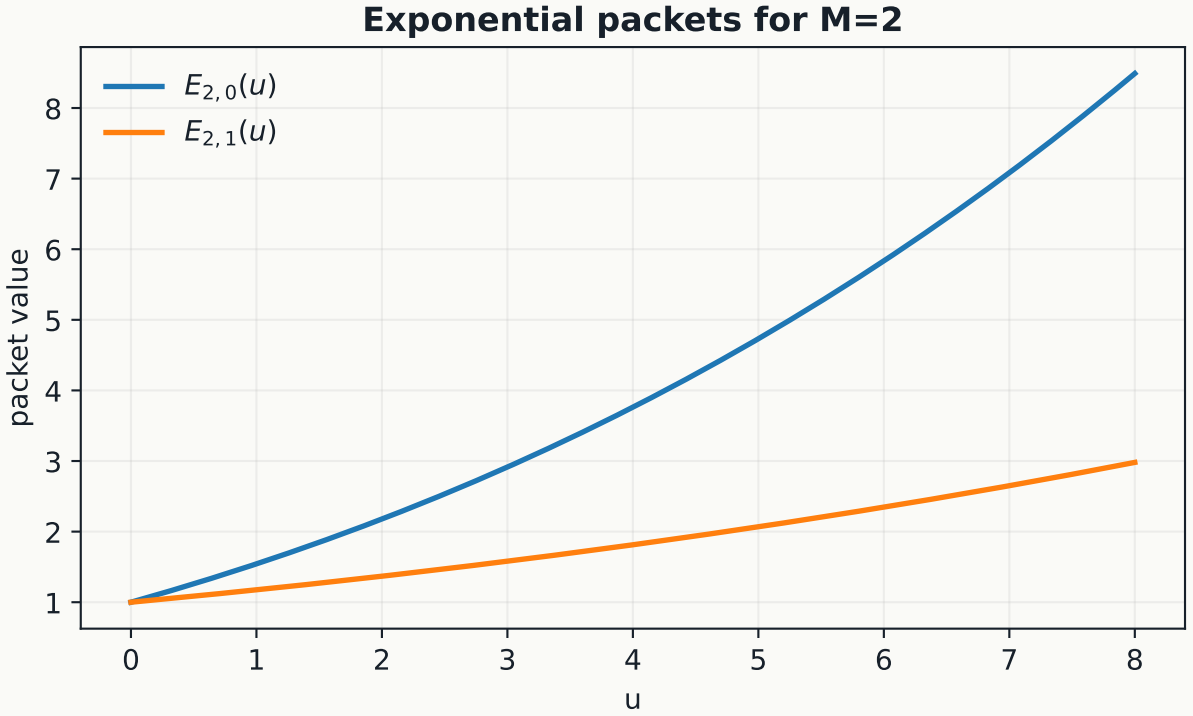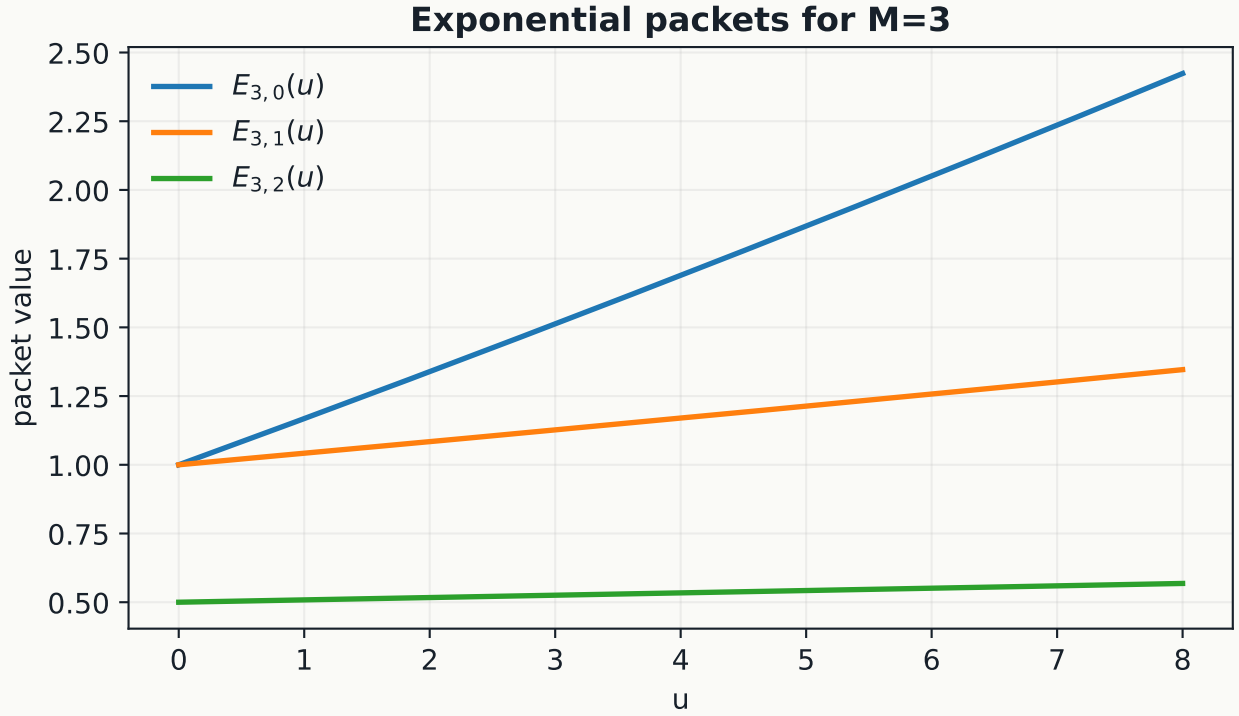The scalar exponential becomes a vector hypergeometric function